

AVERY CHONG

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EDUCATION

University of British Columbia

Vancouver, Canada

Bachelor of Science Co-op, Major in Computer Science

Expected Grad: May 2029

- **Coursework:** Intro to Software Engineering, Relational Databases, Algorithms & Data Structures, Computer Systems, Software Construction, Data Science, Statistics, Applied Machine Learning, Operating Systems
- **Awards:** Outstanding International Student Award; Diversity in CS Hackathon Winner (Matchie)

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java, C/C++, R

Data Engineering: ETL pipelines, Airflow, Docker, GitHub Actions, embedding-based deduplication, data validation, Git

Data Modeling: Relational schema design (BCNF/3NF), normalized multi-table modeling, Row-Level Security

Databases: PostgreSQL, Neon, Supabase, MongoDB, Oracle, SQLite

Frameworks & Libraries: FastAPI, REST APIs, Node.js, Express, Luxon, pandas, scikit-learn, Hugging Face

TECHNICAL PROJECTS

SoCurious (NLP Question Pipeline) | *Python, FastAPI, PostgreSQL, Hugging Face, GitHub Actions*

Aug 2026

- Built the backend for a conversation-starter question app: a multi-source ETL pipeline (**1,484+ posts, 6 subreddits**, RSS) with a **7-rule** cheap-to-expensive filter layer, catching **2** over-aggressive rules before they reached production.
- Benchmarked sentence-embedding deduplication against a fuzzy-matching approach on **525 real questions**, then engineered the pipeline around embeddings after proving **100% recall vs. 66.7%** for fuzzy matching.
- Shipped as a full-stack app, **React/Vite/TypeScript** frontend on **Vercel**, **FastAPI** backend on **Render**, **Neon Postgres**, with **GitHub Actions** orchestrating scheduled pipeline runs and model caching.

blindspot (YouTube Hate-Speech Pipeline) | *Python, Airflow, Docker, Hugging Face, pandas*

Sept 2026 – Present

- Built an **Airflow** pipeline in **Docker** that ingests YouTube comments into a bronze layer, and scored them for **sentiment and emotion** with Hugging Face models.
- Integrating cardiffnlp's **7-class RoBERTa hate classifier** and a **BigQuery star schema**, and documenting its known gaps on covert hate and the disability category.

Eepy (Couple Sleep Accountability App) | *React Native, Node.js, PostgreSQL, Supabase*

May 2026 – Present

- Designed a **UTC-anchored** data model recomputing shared time boundaries from raw per-user schedule data on each check-in event, replacing a fixed-window design that produced inconsistent state across a **26+ hour** timezone gap.
- Restricted database access at the **row level** so users can only touch their own data, across **6 PostgreSQL tables**.
- Eliminated a streak-counting **race condition** with a single **atomic database write**, tested against midnight and daylight-saving edge cases.

Matchie (Diversity in CS Winner) | *React, TypeScript, MongoDB, Clerk, Vercel*

YouCode

- Built a volunteer-matching platform connecting BC students to nonprofits: **semantic embedding search (Nomic Embed v1.5)** with cosine similarity ranking across **4 MongoDB collections**, replacing brittle rule-based filtering.

TravelWrap (School Project) | *TypeScript, Node.js, Express, Oracle*

Jan 2026 – Mar 2026

- Designed fully normalized relational schemas (**BCNF/3NF**) across **15+ tables** for a travel-planning app, modeling travelers, trips, destinations, and purchases, with referential integrity enforced via **cascading foreign keys**.
- Designed the backend API layer, owning end-to-end data design in a **team of 3**.

EXPERIENCE

nwPlus (UX/UI Designer & External Engagement Coordinator)

Sept 2025 – Present

- Contributed to day-of operations and designed website/merch (reaching **1,250+ participants**) for three of Western Canada's largest hackathons, **HackCamp**, **nwHacks**, and **cmd-f**, as part of a **55-member team**.
- Led a **Pitching 101** and **Inclusive Language** workshop for **50+ hackers** and **40+ mentors/judges/volunteers**.